Ambient Polymorphism

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Abstract

…

Introduction

PostgreSQL 11.22 (2023) returns zero for the following query (National Institute of Standards and Technology, n.d.):

NumAcc4.sql

```sql
SELECT stddev(
  CASE
    WHEN k = 1 THEN 10000000.2
    WHEN k % 2 = 0 THEN 10000000.1
    ELSE 10000000.3
  END :: double precision
)
FROM generate_series(1, 1001) AS k;
```

Why? Up until version 12, Postgres computed standard deviation by first computing $a = \sum x$ and $b = \sum x^2$ …

float.c

```c
N += 1.0;
sumX += newval;
CHECKFLOATVAL(sumX, isinf(transvalues[1]) || isinf(newval), true);
sumX2 += newval * newval;
CHECKFLOATVAL(sumX2, isinf(transvalues[2]) || isinf(newval), true);
```

… and then computing $Nb - a^2$:

float.c

```c
numerator = N * sumX2 - sumX * sumX;
CHECKFLOATVAL(numerator, isinf(sumX2) || isinf(sumX), true);

/* Watch out for roundoff error producing a negative numerator */
if (numerator <= 0.0)
    PG_RETURN_FLOAT8(0.0);
```

This catastrophic cancellation is

> the notorious desk top calculator algorithm, also known as the one-pass algorithm.
>
> — Simon and Lesage (1988)

Bibliography

National Institute of Standards and Technology. n.d. “StRD Dataset NumAcc4.” Accessed November 3, 2026. https://www.itl.nist.gov/div898/strd/univ/numacc4.html.

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Simon, Stephen D., and James P. Lesage. 1988. “Benchmarking Numerical Accuracy of Statistical Algorithms.” *Computational Statistics & Data Analysis* 7 (2): 197–209. https://doi.org/10.1016/0167-9473(88)90093-X.